

Chapter 1 非形式邏輯

Informal Logic

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1.1 Premises and Conclusions

前提與結論

Identification of Components 組成要素識別	Premise Indicators 前提指示詞	Conclusion Indicators 結論指示詞
	簡單定義：商販用來引導「生產證據」或「訴苦理由」的開頭關鍵字。 Simple Definition: Trigger words used by merchants to introduce "production evidence" or "reasons for complaining." 日常中招（只認死字）：以為只有看到「因為（Because）」才算理由。 Common Mistake (Relying on Jargon): Thinking a reason only counts if you see the word "because." 鬼才思維（降維打擊）：注意「鑑於（Given that）」、「由於（Since）」、「理由是（On the grounds that）」，這些都是他在搬出他的牛奶檢驗報告。 Genius Insight (Next-Level Up): Look out for phrases like "given that," "since," or "on the grounds that"—these are all signs that he is bringing out his milk test report.	簡單定義：商販鋪墊了很久，最後準備伸手掏你錢包的「收線字眼」。 Simple Definition: The "hook words" used after a long setup, signaling that the merchant is ready to reach into your wallet. 日常中招（只認死字）：以為只有看到「所以（Therefore）」才叫結論。 Common Mistake (Relying on Jargon): Thinking a conclusion only exists when the word "therefore" is used. 鬼才思維（降維打擊）：注意「由此可見（It follows that）」、「這意味著（Which means）」、「這證明了（This proves that）」，這就是他準備把牛奶塞給你的那一刻。 Genius Insight (Next-Level Up): Watch for "it follows that," "which means," or "this proves that"—that is the exact moment he is shoving the milk carton into your hands.

Unstated Elements 未陳述要素	Hidden Assumptions 隱含假設	Enthymemes 省略論證
	簡單定義：藏在肚子裡不說出來，但心裡當作理所當然的「潛規則」。 Simple Definition: The "hidden rule" that is kept in the belly unsaid, but taken for granted in the heart. 正面例子(把話說全)：這盒牛奶包裝膨脹了，因為膨脹的牛奶就是壞了，所以這盒牛奶不能喝。 Positive Example (Fully Stated): This milk carton is bloated, because bloated milk is spoiled, so this milk is undrinkable. 反面例子(藏起假設)：這盒牛奶包裝膨脹了，所以這盒牛奶不能喝。(這時「膨脹等於壞了」就是被藏起來的隱含假設)。 Negative Example (Hiding Assumption): This milk carton is bloated, so this milk is undrinkable. (Here, "bloated equals spoiled" is the hidden hidden assumption). 鬼才思維：這就像相親時對方說「我很有安全感」，他肚子裡沒說的隱含假設是「因為我很胖」。 Genius Insight: It's like a blind date saying "I give off a great sense of security," while his unsaid hidden assumption is "because I am very fat."	簡單定義：故意把話講一半、漏掉一塊拼圖的「殘缺話術」。 Simple Definition: The "broken speech trick" that intentionally tells only half the story, leaving out a piece of the puzzle. 完整論證(兩塊拼圖)：所有過期的牛奶都有毒，你這盒是過期牛奶，所以你這盒有毒。 Complete Argument (Both Pieces): All expired milk is toxic. Your carton is expired milk. Therefore, your carton is toxic. 省略論證(漏掉拼圖)：你這盒牛奶都過期了，你這是在投毒啊！(直接省略了「過期牛奶都有毒」這句大結論，聽起來更有殺傷力)。 Enthymeme (Missing Piece): Your milk is expired, you are poisoning me! (Directly skipping the grand conclusion "all expired milk is toxic" to make it sound more aggressive). 鬼才思維：省略論證就是講話只講一半，剩下的一半讓你猜，高明的騙子最喜歡用這招來帶節奏。 Genius Insight: An enthymeme is speaking only half the truth and making you guess the rest; clever scammers love this trick to control the narrative.

Contextual Analysis 語境分析	Rhetorical Context 修辭語境	Background Beliefs 背景信念
	簡單定義：說話時的現場氣氛、說話人的真正動機，以及聽話人的心理狀態。 Simple Definition: The live vibe of the conversation, the speaker's true motive, and the listener's mental state. 正面分析（看清動機）：黑心老闆在快倒閉的超市裡，一臉流淚地大喊「這批牛奶跳樓大降價」，你分析出他是急著套現逃跑。 Positive Analysis (Reading the Motive): A shady boss inside a failing supermarket cries out "This milk is on a suicidal clearance sale," and you deduce he is desperate to cash out and run. 反面分析（只聽字面）：黑心老闆在快倒閉的超市裡，一臉流淚地大喊「這批牛奶跳樓大降價」，你以為他是在做慈善、流下感動的淚水。 Negative Analysis (Just the Literal Words): A shady boss inside a failing supermarket cries out "This milk is on a suicidal clearance sale," and you genuinely believe he is doing charity and crying tears of joy. 鬼才思維：同樣一句「這牛奶對身體好」，初戀女友送給你叫愛心，推銷員塞給你叫業績。 Genius Insight: The exact same phrase "this milk is good for your body" from your first love is called affection; from a salesman, it's just called hitting his target.	簡單定義：全大眾心裡早就默認、不需要特別解釋的常識與集體認知。 Simple Definition: Common sense and collective understanding that the entire public has already taken for granted without needing an explanation. 正面利用（順水推舟）：老闆在廣告牌上寫「純天然、無添加牧場牛奶」，他成功利用了大家「天然等於健康」的背景信念。 Positive Exploitation (Go with the Flow): The owner writes "pure natural, additive-free farm milk" on the billboard, successfully exploiting the public's background belief that "natural equals healthy." 反面中招（踢到鐵板）：老闆以為可以隨便賣發霉的奶酪，對外宣稱這是新口味，結果低估了全大眾「發霉食物有毒」的集體背景信念。 Negative Mistake (Hitting a Wall): The owner thinks he can easily sell moldy cheese by calling it a new flavor, underestimating the public's collective background belief that "moldy food is toxic." 鬼才思維：背景信念就是大家的腦袋裡早就裝好的軟體，你跟人說話，千萬別去挑戰那個已經設定好的系統。 Genius Insight: Background beliefs are the software pre-installed in everyone's brain; when talking to people, never try to challenge that preset system.

1.2 Argument Analysis

論證分析

Structural Mapping 結構圖解				
	Convergent Structure 收斂結構	Divergent Structure 發散結構	Serial Structure 串聯結構	Linked Structure 並聯結構
Simple Definition 簡單定義	Independent Reasons to One Goal 多個理由各自獨立支持同一個結論。 Like multiple different customers all pointing finger at the same bad shop. 就像多個不同的顧客，各自獨立指著同一家黑店罵。	One Reason to Multiple Conclusions 一個核心理由同時推導出多個不同的結論。 Like a single source of bad milk ruining multiple product brands at once. 就像同一個壞牛奶源頭，同時毀掉多個不同的產品品牌。	Chain Reaction Step by Step 理由像骨牌一樣，一個推導出下一個，最後才到終點。 Like a multi-stage production line where step A leads to B, then B leads to C. 就像多級生產線，步驟 A 導致 B，B 再導向 C，一環扣一環。	Co-dependent Reasons Teamwork 多個理由必須綁在一起同時成立，才能支持結論。 Like two partners collaborating; if you remove either one, the deal collapses. 就像兩個合夥人打配合，只要抽走任何一個，結論就崩塌。
Business Example 商戰例子	Conclusion: This milk shop is a scam. 結論：這家牛奶店是黑店。 Reason 1: Their milk has water inside. 理由一：他們的牛奶裡面摻了水。 Reason 2: Their clerk has a bad attitude. 理由二：他們的店員態度非常惡劣。 (Even if reason 2 is false, reason 1 alone still proves they are a scam.) (即便理由二不成立，理由一單獨也能證明他們是黑店。)	Reason: This milk brand has been found toxic. 理由：這個牛奶品牌被查出有毒。 Conclusion 1: Supermarkets must remove it. 結論一：超市必須馬上將它下架。 Conclusion 2: Stock price of this company will crash. 結論二：這家公司的股價一定會暴跌。 (One single bomb blows up multiple spots at the same time.) (一個地雷同時炸飛多個地方。)	Chain Reaction: 鏈式反應： Reason 1: The cowshed has been infected by bacteria. 理由一：牛棚被細菌感染了。 Leading to Reason 2: The milk produced contains toxic elements. 導向理由二：擠出來的牛奶含有毒素。 Leading to Final Conclusion: Customers who drink it will get sick. 導向最終結論：喝了它的顧客會生病。 (If step one didn't happen, the whole chain wouldn't exist.) (如果第一步沒發生，後面的連鎖反應就不存在。)	Conclusion: The manager must apologize today. 結論：經理今天必須公開道歉。 Reason 1: The store sold expired milk to a kid. 理由一：店裡把過期牛奶賣給了小孩。 Reason 2: The manager previously promised zero tolerance for expired food. 理由二：經理之前承諾過對過期食品零容忍。 (These two reasons must work together to force the apology.) (這兩個理由必須合體，才能逼經理道歉。)

Evaluation Protocols 評估協定		
Acceptability Criteria 可接受性標準	Relevance Assessment 相關性評估	Sufficiency Verification 充分性驗證
簡單定義：第一道門，檢查證據是不是真的。 Simple Definition: Gate 1, checking if the evidence is true. 正常（過關）：這牛奶安全，因為通過了國家安全檢測。 Normal (Passed): This milk is safe because it passed the national safety test. 神經病（翻車）：這牛奶安全，因為我家養的貓用舌頭舔過。 Crazy (Failed): This milk is safe because my pet cat licked it with its tongue. 鬼才思維：地基要是坨屎，後面吹得再好也別聽。 Genius Insight: If the foundation is garbage, don't waste time listening to the rest.	簡單定義：第二道門，檢查證據跟產品有沒有關係。 Simple Definition: Gate 2, checking if the evidence relates to the product. 正常（過關）：你該買這牛奶，因為它含有超高含量的鈣。 Normal (Passed): You should buy this milk because it contains very high calcium. 神經病（翻車）：你該買這牛奶，因為我們請的代言人長得超帥。 Crazy (Failed): You should buy this milk because our spokesperson is super handsome. 鬼才思維：別被厲害但無關的話轉移了你的視線。 Genius Insight: Don't let impressive but irrelevant words distract you.	簡單定義：第三道門，檢查證據數量夠不夠鎖定勝局。 Simple Definition: Gate 3, checking if the amount of evidence secures victory. 正常（過關）：牛奶沒壞，因為檢驗報告、出廠日期、冷鏈紀錄全都有。 Normal (Passed): The milk is good because we have the lab report, date stamp, and cold chain logs. 神經病（翻車）：牛奶沒壞，因為我剛剛喝了一口，現在肚子還沒痛。 Crazy (Failed): The milk is good because I just took a sip and my stomach doesn't hurt yet. 鬼才思維：拿一粒米就想證明請吃了大餐，那是耍流氓。 Genius Insight: Trying to prove a feast with a single grain of rice is pure gaslighting.

1.3 Argument Reconstruction 論證重建

Interpretative Principles 解釋原則		
	Principle of Charity 寬容原則	Principle of Fidelity 忠實原則
Simple Definition 簡單定義	Turn Opponent's Words into Their Strongest Version 把對手說的話，重新包裝成最聰明、最無懈可擊的版本來解讀。 Treating your opponent like a master, not an idiot, to avoid fighting a straw man. 把對手當成高手而不是白痴來解讀，不自己扎個稻草人來打。	Keep Opponent's Real Intentions Unchanged 嚴格扣緊對手說話時的本意和語境，不隨便歪曲或加料。 Treating your opponent like an honest witness; do not twist their words to frame them. 把對手當成誠實的證人，不隨便斷章取義或栽贓。
Business Context 商戰語境	The customer says wildly: "Your milk expired yesterday! I will sue your shop until it goes bankrupt!" 顧客暴怒大喊：「你們的牛奶昨天過期了！我要告到你們店倒閉！」	The merchant says carefully: "Our milk has some chemical elements to keep it fresh, but it's safe." 商販謹慎地說：「我們的牛奶加了一點化學成分保鮮，但絕對安全。」
Reconstructed Form 重建後的解讀	Passed (Charitable Interpretation): 過關(寬容解讀) "The customer is worried about food safety risks and wants us to take responsibility." 「顧客是擔心食品安全風險，希望我們承諾負責。」 Failed (Uncharitable Interpretation): 翻車(不寬容解讀) "The customer is a crazy person who wants to destroy the retail industry." 「這個顧客是個想毀滅整個零售業的精神病。」	Passed (Faithful Interpretation): 過關(忠實解讀) "The merchant admits to using legal chemical preservatives to extend shelf life." 「商販承認使用了合法的化學防腐劑來延長保鮮期。」 Failed (Unfaithful Interpretation): 翻車(不忠實解讀) "The merchant just openly admitted that he is putting poison inside the milk!" 「這商販剛剛親口承認他在牛奶裡面下毒！」

Textual Standardization 文本標準化		
1 Elimination of Irrelevancies 消除無關訊息	2 Disambiguation of Terms 消除術語歧義	3 Logical Form Formulating 邏輯形式公式化
簡單定義：像用篩子隔渣一樣，把對手的髒話、廢話、眼淚全部倒掉，只留下有用的邏輯證據。 Simple Definition: Like filtering dregs with a sieve, dump all the cursed words, nonsense, and tears from your opponent, leaving only useful logical evidence. 商販胡扯：天啊！我們全家經營這家牛奶店三十年了，每天起得比雞早，我老婆還因為送奶摔斷了腿，你怎麼能因為一盒過期牛奶就懷疑我們的良心？ Merchant's Nonsense: Oh my god! Our family has run this milk shop for thirty years, waking up earlier than roosters, and my wife even broke her leg delivering milk! How can you doubt our conscience just because of one expired milk carton? 過濾濾鏡（只留骨架）：這家店賣出了一盒過期牛奶。 X-Ray Filter (Skeleton Only): This shop sold a carton of expired milk.	簡單定義：把對手說得不清不楚、故意一詞多義的「迷霧彈」字眼，徹底鎖定成同一個死定義。 Simple Definition: Taking the vague, double-meaning words your opponent uses as smoke bombs, and locking them down into a single, unchangeable definition. 商販迷霧：我們的牛奶絕對是「純天然」的，所以喝了絕對對身體「好」。 Merchant's Smoke: Our milk is absolutely "pure natural," so drinking it is definitely "good" for your body. 消除迷霧（鎖定定義）：「純天然」指的是從牛身上擠出來、未加防腐劑；「好」指的是能提供人體所需的鈣質。 Clearing the Smoke (Locking Definition): "Pure natural" means squeezed from a cow without preservatives; "good" means providing the calcium required by the human body.	簡單定義：把過濾乾淨、鎖定定義的話，排成「一、二、三，所以四」的流水線，讓他無法耍賴。 Simple Definition: Arranging the filtered, locked-down words into a neat conveyor belt of "One, Two, Three, therefore Four," giving them no way to wiggle out. 混亂原始對話：你這店賣過期奶，小孩喝了肯定拉肚子，你今天必須賠錢！ Messy Raw Speech: Your shop sells expired milk, kids will definitely get diarrhea if they drink it, you must pay me back today! 公式化重建： Formulated Reconstruction: 前提一：這家店賣出了過期牛奶。 Premise 1: This shop sold expired milk. 前提二：喝過期牛奶會導致小孩拉肚子。 Premise 2: Drinking expired milk causes kids to get diarrhea. 結論：這家店應該對拉肚子的後果進行賠償。 Conclusion: This shop should pay compensation for the diarrhea consequence.

1.4 Logical Fallacies

邏輯謬誤

第一優先：煙霧彈流派（轉移視線） Priority 1: The Smoke Bomb School (Distraction)	1. 人身攻擊 (Ad Hominem)	人話定義：打不過你的證據，就改為攻擊你的私德。 Simple Definition: Can't beat your evidence, so they attack your personal life instead. 鬼才記憶：你罵他牛奶壞，他罵你人品壞。
	2. 紅鯡魚 (Red Herring)	人話定義：突然扯一個完全不相干的事，帶全場觀眾集體跑題。 Simple Definition: Crashing in with a totally unrelated topic to make everyone lose the plot. 鬼才記憶：發現牛奶長蟲，他突然大喊「看！那邊有人發錢！」
	3. 訴諸情感 (Appeal to Emotion)	人話定義：講不出道理，就當眾哭可憐或煽動恐懼來情緒勒索。 Simple Definition: Failing to reason, so they cry or trigger fear for emotional blackmail. 鬼才記憶：用眼淚和奶奶的哭聲，去感化你胃裡的細菌。
	4. 稻草人 (Straw Man)	人話定義：把你的話極端化、弱智化，編成一個傻子觀點來打。 Simple Definition: Exaggerating your words into a stupid version just to beat it down easily. 鬼才記憶：你說「糖太多」，他扭頭大喊「他想讓我們的小孩餓死！」
第二優先：神棍吹牛流派（偽造證據） Priority 2: The Charlatan School (Fake Evidence)	5. 訴諸不當權威 (Appeal to Unqualified Authority)	人話定義：請外行的名人來為專家領域的產品瞎背書。 Simple Definition: Getting a famous outsider to blindly endorse a specialized expert product. 鬼才記憶：找個游泳冠軍來向你保證這盒奶能治高血壓。
	6. 虛假因果 (False Cause)	人話定義：僅僅因為兩件事先後發生，就硬生生綁上因果關係。 Simple Definition: Forcing a cause-and-effect bond just because two things happened one after the other. 鬼才記憶：喝完牛奶車胎爆了，他就說牛肚子裡有詛咒。
	7. 輕率概括 (Hasty Generalization)	人話定義：樣本太少，只拿一兩個極端個案就直接亂下大結論。 Simple Definition: Using way too little data to jump into a massive, sweeping conclusion.

		鬼才記憶：一個員工喝了沒拉肚子，就宣布這批過期奶是神仙水。
	8. 訴諸無知 (Appeal to Ignorance)	人話定義：把「還沒被科學證明是錯的」直接等同於「它就是對的」。 Simple Definition: Treating "not yet proven false" as absolute proof that "it is true." 鬼才記憶：科學家還沒證明我的奶有毒，那它就是安全的。
第三優先：強盜邏輯流派(霸王條款) Priority 3: The Highwayman School (Forced Trap)	9. 假兩難 (False Dilemma)	人話定義：明明路有很多條，卻故意恐嚇你只有兩個極端選擇。 Simple Definition: Pretending there are only two extreme paths when plenty of choices exist. 鬼才記憶：不買天價奶，你就是想看著孩子骨折的狠心父母。
	10. 複合問題 (Complex Question)	人話定義：在問題裡偷偷埋一個陷阱，不管你答是或不是都直接認罪。 Simple Definition: Hiding a trap inside a question so any answer automatically confesses to a crime. 鬼才記憶：溫柔地問你：「你從哪天起停止往牛奶裡倒馬桶水的？」
	11. 乞題 (Begging the Question)	人話定義：把結論換個說法，直接當成支持這個結論的理由。 Simple Definition: Rephrasing the conclusion and using it as the premise to prove itself. 鬼才記憶：我的奶最好因為它超越一切；為什麼超越一切？因為它最好。
第四優先：魔術師流派(文字障眼法) Priority 4: The Magician School (Word Illusions)	12. 偷換概念 (Equivocation)	人話定義：同一個詞，在中途悄悄換成另一個完全不同的意思。 Simple Definition: Silently mutating the meaning of a single word halfway through the talk. 鬼才記憶：簽合同說用「純」潔的心做奶，交貨時裡面全是「純」水。
	13. 合成與分割 (Composition / Division)	人話定義：瞎套整體與局部的關係，以為局部安全等於整體安全(合成)，或整體厲害等於每個人都厲害(分割)。 Simple Definition: Blindly assuming parts equal the whole (Composition) or the whole equals the parts (Division). 鬼才記憶：每個原子安全不等於喝十噸安全；公司頂級不等於送奶工是神仙。
	14. 語法歧義 (Amphiboly)	人話定義：句子語法結構有漏洞，導致一句話可以讀出兩種完全相反的意思。 Simple Definition: A sentence structure so messy that it can be read in two completely opposite ways. 鬼才記憶：「賣牛奶給帶有裂殼的顧客」，留個語法黑洞等律師來鑽。